👩🏾‍🏫 Staff Guide · Small Engines Overhaul · Unit 1 Measuring Tools (Level 1)

Level 1 = about 2nd-grade reading, with emoji and pictures. SLE students stay in the class teacher's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: “Week 1: Measurement Lab Sheet (dial calipers, micrometer, stations), Micrometer Handout, Measuring cylinder bore video”)	**Level 1 piece**
Measurement Lab Sheet (4 stations)	Measurement Stations lab packet + notes A, B
Micrometer Handout (scanned PDF)	Slides “Reading a Micrometer” + notes B
Measuring cylinder bore video	Video guide + slides “Measuring the Cylinder Bore”
Class slides	Deck: 14 slides, Words to Know first, “point to the picture” checks, recap
His worksheets (grouped into lessons)	Guided notes: 15 items + key
Class quiz / notes check	Quiz: 8 questions + key
Vocabulary	8 picture cards (cut apart)
—	1 game on the class site: 10 questions + a matching round

2. Supports built in

- **Read aloud** every line; student may point, say, or circle (point = full answer)
- One idea per slide; big pictures; 2–3 picture choices
- Digital calipers (if available) for students who cannot read a dial; same points
- Calculator for adding bar + dial
- Number line card for thousandths

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the “look for” part
- Cover one choice to make it 2

4. What to grade

- Measurement Lab: L1 picks the right tool and an adult reads the number; L2 reads a digital or dial caliper; L3 reads a micrometer with a frame
- **Notes**: same points as his worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm in his files say “check with the class teacher”.

🚧 Under construction (check with the class teacher)

- **Micrometer Handout**: The handout is a scanned image with no text; our notes teach the same skill.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

👩‍🏫 Staff Guide · Measurement Stations (Level 1)

Small Engines Overhaul · Unit 1 · goes with the class “Measurement Lab Sheet” (lab points). Level 1 = about 2nd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Task card**: Modified from the class “Measurement Lab Sheet”. 7 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 3 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- Show me the first step.
- Which tool do we need now?
- Where do your hands go while the tool is turning?
- What should we do when we finish?

3. Modifications

- Model first, then the student tries with adult supervision
- Picture task card at the bench
- Parts tray with labeled cups for small parts

4. What to grade (lab points)

- Did the steps with support (photo proof)
- My Check page done
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver (SLE).

🚧 Under construction (check with the class teacher)

- **This packet**: Station specs depend on the engines in Mr. Davis's shop (the Lab Sheet leaves the model blank); staff fill in the spec column.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

📏 Measuring Tools Notes

Small Engines Overhaul · Unit 1 · Level 1

Name: _____

✏️ A. Trace the word. Then copy it.



ruler _____



caliper _____



micrometer _____



spec _____

🖼️ B. Fill in the word. Use the pictures.

5.  A _____ measures length.

6.  A _____ has a dial.

7.  A _____ measures tiny sizes.

8.  The _____ is the right size.

Word bank:  **caliper**  **micrometer**  **ruler**  **spec**

⭕ C. Circle the right picture.

9. Before measuring I…



Hit the part



Wipe the part

10. A dropped micrometer…



Gets stronger



Can break

11. 0.001 is…



Very small



Very big

🔗 D. Draw a line from the picture to the word.



caliper



micrometer



feeler gauge



ruler

📖 Measuring Tools: Vocabulary Picture Cards (Level 1)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.



Ruler

measures length



Dial caliper

a tool with a dial that measures size



Micrometer

measures very small sizes



Feeler gauge

thin metal strips

🎯

Specification

the right size from the book

🔢

Thousandth

a very tiny bit (0.001)

⭕

Diameter

distance across a circle



Cylinder bore

the hole the piston moves in

📏 Measurement Stations

Small Engines Overhaul · Unit 1 · Level 1



Name: _____

Modified from the class “Measurement Lab Sheet”.

🧺 Supplies: get it. Check ☐.

- ☐  **Dial caliper** or digital
- ☐  **Micrometer** 0-1 inch
- ☐  **Feeler gauge set**
- ☐  **Ruler**
- ☐  **Rod and crankshaft** station parts
- ☐  **Clean rag**

📏 Do one box. Check ☐ it.

1		Glasses on.	☐
2		Wipe the part.	☐
3		Point to the right tool.	☐
4		Close the jaws with an adult.	☐
5		Read or show the number.	☐
6		Adult helps me write it.	☐
7		Tool back in the case.	☐

🚧 **Under construction:** Station specs depend on the engines in Mr. Davis's shop (the Lab Sheet leaves the model blank); staff fill in the spec column. *Check with the class teacher.*

📏 Measurement Stations · Get Ready

Small Engines Overhaul · Unit 1 · Level 1



Name: _____

👥 My job. Circle it.



Mechanic
fixes



Tool helper
gets tools



Recorder
writes



Clean-up captain
cleans

🧰 Get the tools. Check ☐.

☐  Dial caliper

☐  Micrometer

☐  Feeler gauge set

☐  Ruler

☐  Rod and crankshaft

☐  Clean rag

⚠️ Be safe!

  Glasses on.

  Coveralls on.

  Tools with an adult.

  Gentle hands.

  Clean up, wash hands.

✅ Before we start

☐ Safety glasses ☐ Coveralls ☐ Adult at bench

📏 Measurement Stations · My Check

Small Engines Overhaul · Unit 1 · Level 1



Name: _____

📸 My photos. Check ☐.

- ☐ 📸 Before
- ☐ 📸 During
- ☐ 📸 After

✅ I did it! Check ☐.

- ☐ 🥽 I wore my safety gear
- ☐ 🔧 I did the engine job
- ☐ 🧽 I cleaned up
- ☐ 🤝 I helped my group

⭕ Circle one face.

🥽 **I was safe.**	😀	🙂	😕
🔧 **I did my job.**	😀	🙂	😕
⭐ **I liked it.**	😀	🙂	😕

⭐ I learned ____. Draw or write.

✅ Measuring Tools Quiz

Small Engines Overhaul · Unit 1 · Level 1 · 8 questions. Circle the picture.

Name: _____

1. 📏 Measures length…

- 🧹 Broom
- 📏 Ruler

2. 🗜️ Has a dial…

-  Dial caliper
- 🔨 Hammer

3. 🔩 Most exact tool…

-  Micrometer
-  Ruler

4. 📄 Checks a gap…

-  Screwdriver
-  Feeler gauge

5. 🔢 0.001 is…

-  Very big
- 🤏 Very small

6. 🎯 The right size is the…

-  Spec
-  Snack

7. 🕳️ The piston moves in the…

-  Gas tank
-  Cylinder bore

8. 📦 After using a tool…

-  Put it in its case
-  Throw it

🎬 Measuring Cylinder Bore

Small Engines Overhaul · Unit 1 · Level 1 video guide · class points pts

Name: _____

👀 Watch for these pictures.



YouTube video linked in Canvas: https://www.youtube.com/watch?v=6jsiVBDfqz4

1. The piston moves in the…

-  **Cylinder**
-  **Gas tank**

2. They measure with a…

-  **Bore gauge or micrometer**
-  **Hammer**

3. They measure…

-  **Only once**
-  **In more than one spot**

4. Then they compare to the…

-  **Menu**
-  **Spec**

⭐ **My favorite part:** _____

📏 Measuring Tools Notes · KEY

Small Engines Overhaul · Unit 1 · Level 1

Name: _____

✏️ A. Trace the word. Then copy it.

 ruler _____

 caliper _____

 micrometer _____

 spec _____

🖼️ B. Fill in the word. Use the pictures.

5.  A **ruler** measures length.

6.  A **caliper** has a dial.

7.  A **micrometer** measures tiny sizes.

8.  The **spec** is the right size.

Word bank:  **caliper**  **micrometer**  **ruler**  **spec**

⭕ C. Circle the right picture.

9. Before measuring I…

- 🔨 Hit the part
- 🧽 **Wipe the part** ✓

10. A dropped micrometer…

- 💪 Gets stronger
- 💔 **Can break** ✓

11. 0.001 is…

- 🤏 **Very small** ✓
- 🏔️ Very big

🔗 D. Draw a line from the picture to the word.

Key: 📏 → ruler · 🗜️ → caliper · 🔩 → micrometer · 📄 → feeler gauge

	**caliper**
	**micrometer**
	**feeler gauge**
	**ruler**

🔑 *Key written by our team from the class teacher's materials, please check.*

✅ Measuring Tools Quiz · KEY

Small Engines Overhaul · Unit 1 · Level 1 · 8 questions. Circle the picture.

Name: _____

1. 📏 Measures length… Measurement Lab

- Broom
- **Ruler** ✓

2. 🗜️ Has a dial… Measurement Lab

- **Dial caliper** ✓
- Hammer

3. 🔩 Most exact tool… Micrometer Handout

- **Micrometer** ✓
- Ruler

4. 📄 Checks a gap… Measurement Lab

- Screwdriver
- **Feeler gauge** ✓

5. 🔢 0.001 is… Notes

- Very big
- **Very small** ✓

6. 🎯 The right size is the… Notes

- [x] 🎯 Spec
- [ ] 🍕 Snack

7. 🕳️ The piston moves in the… Cylinder bore video

- [ ]  Gas tank
- [x]  Cylinder bore

8. 📦 After using a tool… Notes

- [x] 📦 Put it in its case
- [ ] 🗑️ Throw it

Answers

1-B (Ruler) · 2-A (Dial caliper) · 3-A (Micrometer) · 4-B (Feeler gauge) · 5-B (Very small) · 6-A (Spec) · 7-B (Cylinder bore) · 8-A (Put it in its case)

Level 1 version of the gen ed “Measurement Lab Sheet”. Read every question aloud. Score: __ / 8.

🔑 *Key written by our team from the class teacher's materials, please check.*

🎬 Measuring Cylinder Bore · KEY

Small Engines Overhaul · Unit 1 · Level 1 video guide · class points pts

Name: _____

👀 Watch for these pictures.



YouTube video linked in Canvas: https://www.youtube.com/watch?v=6jsiVBDfqz4

1. The piston moves in the…

- [x] Cylinder
- [ ] Gas tank

2. They measure with a…

- [x] Bore gauge or micrometer
- [ ] Hammer

3. They measure…

- [ ] Only once
- [x] In more than one spot

4. Then they compare to the…

- [ ] Menu
- [x] Spec

⭐ **My favorite part:** _____

Goes with the class “Measuring cylinder bore (video notes)” (class points pts). Same points when complete.
🔑 Key written by our team from the class teacher's materials, please check.